

### 13.1:

Arm has a fixed base, so do not need parameters for the base position and orientation, but only for the joints. Each one of the joints will then be attributed to one degree of freedom.

Parameters:  $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6, d_7)$

$\theta_1$ – $\theta_6$  are the rotation angles of the six joints and  $d_7$  is sliding distance of the final joint.

Configuration space is then  $S^1 \times S^1 \times S^1 \times S^1 \times S^1 \times S^1 \times \mathbb{R}$

Which is simplified to  $(S^1)^6 \times \mathbb{R}$

Each of the revolute joints gives one angular parameter, and the prismatic joint gives one linear parameter. Therefore, the dimension of the configuration space must be 7.

### 13.2

Center node is only used as a connector inside each trapezoid, so it can be replaced with one direct edge across the trapezoid.

In original roadmap, put one node in the center of each trapezoid, one node in the middle of each vertical extension, and connects nodes to boundary nodes by straight line arcs.

The center node can be removed.

For each trapezoid  $\Delta$  in  $T(C_{\text{free}})$ :

1. Only keep the nodes on the vertical walls
2. Let  $u_\Delta$  be the node on the left vertical wall of  $\Delta$  and let  $v_\Delta$  be the node on the right vertical wall of  $\Delta$
3. Add  $(u_\Delta, v_\Delta)$  as an edge.

Draw the edge as a straight segment inside  $\Delta$ . This now replaces the two edges  $u_\Delta \rightarrow \text{center}$  and  $\text{center} \rightarrow v_\Delta$  with just one edge:  $u_\Delta \rightarrow v_\Delta$ . So the graph is smaller and there are less edges.

Completeness:

This will work because trapezoids are convex and the segment between  $u_\Delta$  and  $v_\Delta$  will remain inside the trapezoid. Also the trapezoid lies in  $C_{\text{free}}$ , so it is collision free. Neighboring trapezoids share vertical wall node, so if real free path crosses trapezoids, then changed graph can still move from trapezoid to trapezoid through the wall nodes.

Query Algorithm: Querying from  $p_{\text{start}}$  to  $p_{\text{finish}}$

1. Find  $\Delta_{\text{start}}$  and  $\Delta_{\text{finish}}$  trapezoids
2. If the same, then return the straight segment  $p_{\text{start}}p_{\text{finish}}$
3. If not, then connect  $p_{\text{start}}$  to the wall nodes of  $\Delta_{\text{start}}$
4. Connect  $p_{\text{finish}}$  to the wall nodes of  $\Delta_{\text{finish}}$
5. Run BFS in the change roadmap

6. Return the path  $p_{\text{start}} \rightarrow \text{wall node} \rightarrow \text{roadmap path} \rightarrow \text{wall node} \rightarrow p_{\text{finish}}$

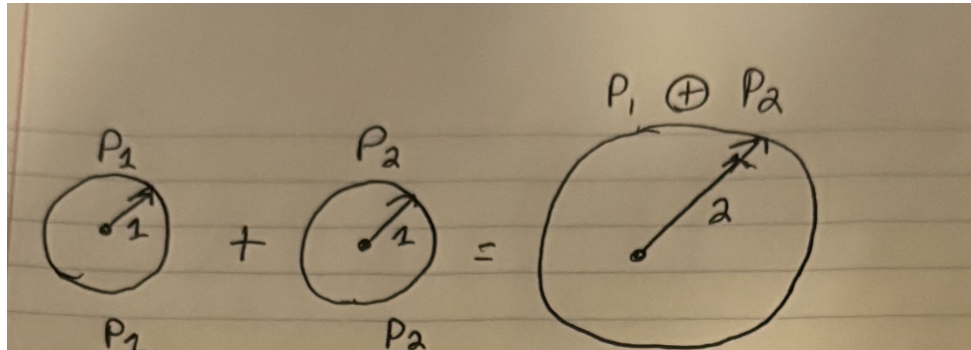
### 13.4

Assume that unit disc has radius of 1, the unit square has side length of 1, and the shapes are in the standard positions. (just assuming from lectures)

A) Both  $P_1$  and  $P_2$  are unit discs

Disc radius 1 + disc radius 1 = disc radius 2

This results in a disc shape with radius 2, so a disc twice as large

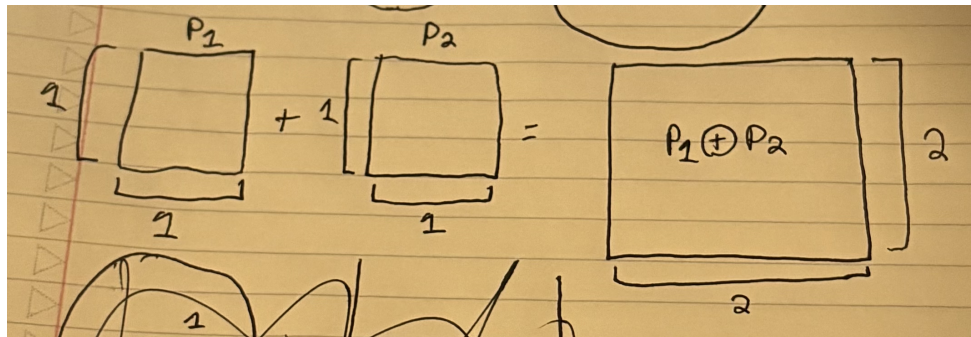


B) Both  $P_1$  and  $P_2$  are unit squares

A unit square plus a unit square gives a square of side length 2

If each square is  $[1, 0] \times [1, 0]$ , then  $[1, 0]^2 \oplus [1, 0]^2 = [2, 0]^2$

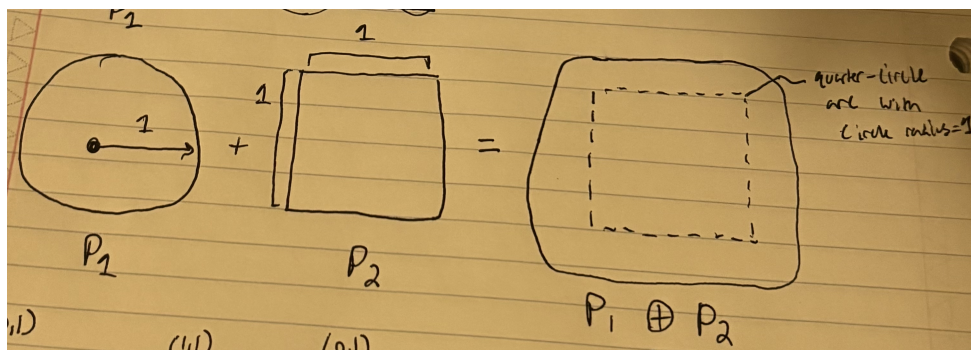
This results in a square shape with side length of 2



C)  $P_1$  is unit disc and  $P_2$  is a unit square

This will result in a square whose distance is increased by 1 in every direction

So it is a rounded square shape whose boundary has 4 straight segments parallel to the square's sides, and 4 quarter-circles of radius 1 at the corners.



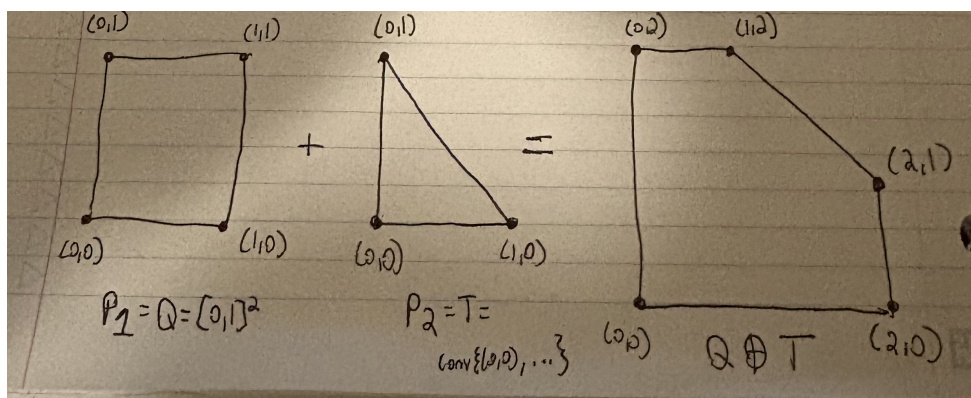
D)  $P_1$  is a unit square and  $P_2$  is triangle with vertices  $(0,0)$ ,  $(1,0)$ ,  $(0,1)$

Square  $Q = [0,1]^2$  and triangle  $T = \text{conv}\{(0,0), (1,0), (0,1)\}$

Minkowski sum will be the convex hull of all of the pairwise sums of the vertices.

This equates to a polygon with vertices:  $(0,0)$ ,  $(2,0)$ ,  $(2,1)$ ,  $(1,2)$ ,  $(0,2)$

Which therefore has a shape of a pentagon.



### 13.6

Need to utilize the dot product definition of extreme in direction  $d$ .

Where  $p$  is extreme on  $P$  in direction  $d$  when  $p \cdot d \geq q \cdot d$  for all  $q \in P$

Minkowski sum is defined as the sums  $p + r$ , with  $p \in P$  and  $r \in R$

Proof:

Let  $p$  be the extreme point of  $P$  in direction  $d$ , and  $r$  is an extreme point of  $R$  in direction  $d$ .

First need to show that  $p + r$  is extreme on  $C_p = P \oplus R$

Take any point  $s \in P \oplus R$

Through the definition of Minkowski sum,  $s = q + t$  for some  $q \in P$  and  $t \in R$

Since  $p$  is extreme on  $P$ ,  $q \cdot d \leq p \cdot d$

Since  $r$  is extreme on  $R$ ,  $t \cdot d \leq r \cdot d$

Therefore,  $s \cdot d = (q + t) \cdot d = q \cdot d + t \cdot d \leq p \cdot d + r \cdot d = (p + r) \cdot d$

So every point  $s$  in  $P \oplus R$  has a dot product at most  $(p + r) \cdot d$

Therefore,  $p + r$  is an extreme point of  $P \oplus R$  in direction  $d$  and the observation in 13.4 is proven.

### 15.1

Need to the theorem: A shortest path among disjoint polygonal obstacles is a polygonal path whose inner vertices are obstacle vertices.

Prove  $O(n)$  Bound:

Let  $S$  have  $n$  obstacle edges. Since the obstacles are simple polygons, they also have  $n$  total obstacle vertices.

Through the theorem above, any shortest path from  $p_{\text{start}}$  to goal is polygonal, and every inner vertex of the path is a vertex of some obstacle. So the path can only turn at obstacle vertices.

- Shortest path can't visit same obstacle vertex twice, as it would not be the shortest path.

So the shortest path has  $n$  inner vertices at most. A polygonal path with  $k$  inner vertices has  $k + 1$  segments, so the number of segments is  $n + 1$  at most, which is in  $O(n)$  time.

For any start and end position, the shortest path has  $O(n)$  segments.

Example:

Place thin triangular/rectangular obstacles between  $p_{\text{start}}$  and goal, switching from top to bottom so that the line has a zig-zag shape.

- $M$  obstacles where the shortest route weaves up and down between them and touches one obstacle vertex at each of its turns.

Since there are  $m$  obstacles, the path has  $\Theta(m)$  bends and therefore  $\Theta(m)$  segments.

Since each obstacle has constant complexity, the total number of edges is equal to  $n = \Theta(m)$ .

So the shortest path has  $\Theta(n)$  segments.